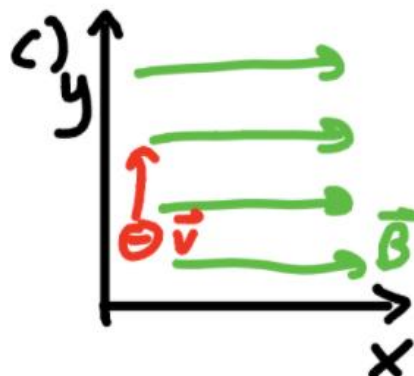
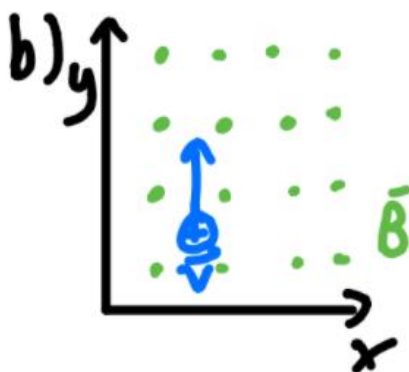
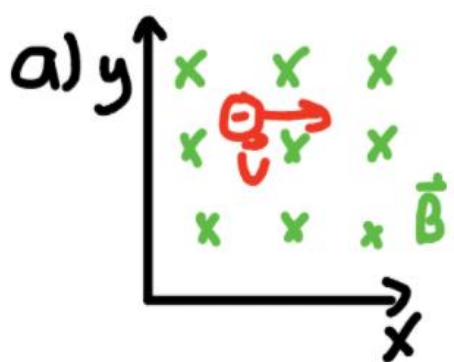
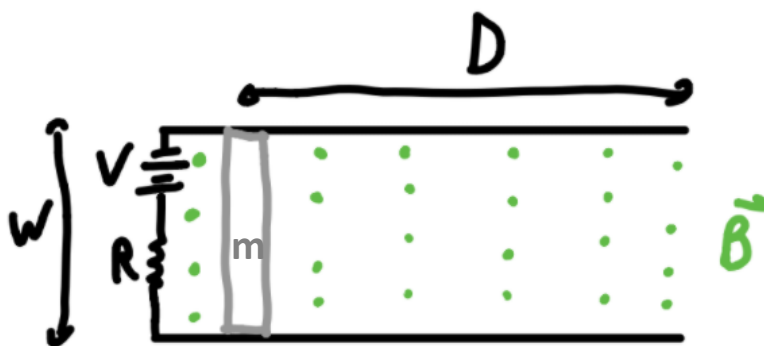


1. For each of the following, give the direction of the force on the particle.



2. A particle (mass = 2.0 mg, charge = $-6.0 \mu\text{C}$) moves in the positive direction along the x axis with a velocity of 3000.0 m/s. It enters a magnetic field of $\vec{B} = (2.0\hat{i} + 3.0\hat{j} + 4.0\hat{k})$ T. What is the magnitude of the acceleration of the particle?



3. A metal bar of mass $m = 0.8 \text{ kg}$ and length $w = 2.0 \text{ m}$ is placed on a rail that is connected to a circuit as shown. The circuit has a battery of $V = 20 \text{ V}$ and a resistor of $R = 40 \Omega$. All other wires have zero internal resistance. This entire strange thing is placed in a magnetic field that is directed out of this page, with a magnitude of $B = 3.0 \text{ T}$.

The bar is released at the point shown with an initial speed of $v_0 = 1.0 \text{ m/s}$. Find the speed of the bar after it travels a distance of $D = 2.0 \text{ m}$.